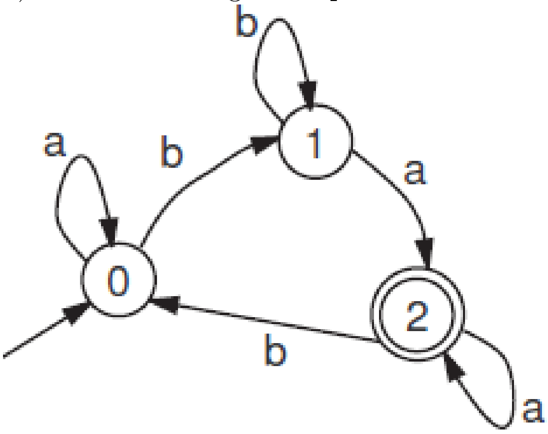
Ministry of Higher Education and Scientific Research
UNIVERSITY M'Hamed BOUGARA – BOUMERDES
FACULTY OF SCIENCE - DEPARTMENT OF COMPUTER SCIENCE

Module : Theory of Languages
Field : L2
Academic Year : 2024-2025

TUTORIAL N 4

Exercise 01

a) Determine a regular expression for the following automaton :



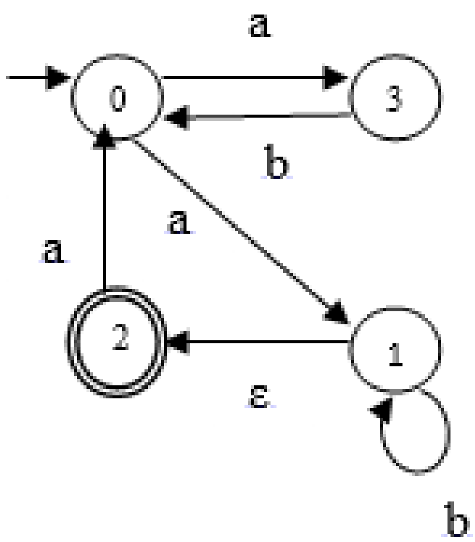
b) Construct the finite automata corresponding to the following regular expressions :
— 10+(0+11)0*1
— (00+01)* + (10+01)*
c) Propose a finite-state automaton and a regular expression for the following languages :
— L1=$\{a^{2n+2}b^pc^{m+1}, n, m, p \geq 0\}$
— L2=$\{wc^{2n+1}, n \geq 1 \; and \; w \in \{a, b\}^* \; and \; |w| = 3m + 1, m \geq 0\}$

Exercise 02

Consider the following two languages :
L1={w $\in \{0, 1\}^*$, such that in every word w of L1, every substring "11" is immediately followed by at least one "0"}
L2={w $\in \{0, 1\}^*$, such that every w contains at least the subsequence "11"}
— Determine a minimal finite-state automaton that accepts language L1.
— Determine a minimal finite-state automaton that accepts language L2.
— Determine a minimal finite-state automaton that accepts the language $L1 \cup L2$.
— Provide a regular expression that denotes the language $L1 \cup L2$.

Exercise 03

1. Given the following nondeterministic automaton M :



— Construct a minimal deterministic automaton M' equivalent to M.
— Determine a right-regular grammar G1 that generates L(M).
2. Given the grammar G2= ({a,b}, {S,A,B}, S, R) with
R={S→abS/aA,
A→bA/B
B→aS/ϵ}
— What is the type of G2 and L(G2) ?
— Compare L(G1) and L(G2).
— Determine a left-regular grammar G3 that generates L(M).

Exercise 04

Among the following languages, which ones are regular ?
— L1= {s =a^n : n is a prime number}
— L2=$\{a^nb^{2m}, n, m >= 0\} \cup \{a^n, n >= 0\}$
— L3= $\{s = a^nb^{2n}, n <= 100\}$
— The set of words having an equal number of zeros and ones.
— The set of words over {0,1} that do not contain three consecutive zeros.

Exercise 05

Provide an extended regular expression describing the following languages :
— All words over {a, b} where each 'a' is preceded by a 'b'.
— All words over {a, b} containing both the factors "aa" and "bb".
— All words over {a, b} containing either "aa" or "bb" but not both.
— All words over {a, b} that do not contain two consecutive 'a's.
— All words over {a, b, c} where the number of 'a's is a multiple of 2.
— All integers (in base ten) that are multiples of 5.

Exercise 06

Consider the following grammar G= ({a, b},{S, A, B, C}, S, R) with R :
R={S->Sa/ Aa/Ca
A->Bb
B->Ca/ Sa/ Aa
C->ϵ }

- Determine a minimal finite-state automaton that accepts the language generated by this grammar.
- Provide a regular expression (denoted EXP1) that represents the language generated by this grammar (denoted L(EXP1)).
- Given the regular expression EXP2= ((a+ab)*+(a+aa)*)*. The language denoted by this expression is L(EXP2). Determine a minimal finite-state automaton that accepts the language denoted by this expression.
- Compare L(EXP1) and L(EXP2) : Do we have L(EXP1)=L(EXP2) or $L(EXP1) \subset L(EXP2)$ or $L(EXP1) \supset L(EXP2)$? Justify your answer.